Fiscal Year:	FY 2024	**Task Last Updated:**	FY 11/09/2023
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per NSSC information (Ed., 8/6/24) NOTE: End date changed to 09/30/2024 per NSSC information (Ed., 11/9/23) NOTE: End date changed to 09/30/2023 per C. Ribeiro/JSC and NSSC information (Ed., 4/7/22) NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 8/13/21) NOTE: End date changed to 09/30/2021 per NSSC information (Ed., 5/20/21) NOTE: End date changed to 02/28/2021 per NSSC information (Ed., 9/21/20) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		

Grant/Contract No.:	80NSSC18K0310
Performance Goal No.:	
Performance Goal Text:	
Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	We have participated in a spaceflight study (MHU-8) to evaluate mouse retina and optic nerve head changes following spaceflight. The objectives of the MHU project are to characterize the impact of spaceflight on mouse neurovascular remodeling and blood-retinal barrier (BRB) integrity in the retina and to study the role of oxidative stress in regulating spaceflight condition-induced cellular damage and cell-cell interaction. Furthermore, we will elucidate the impact of artificial gravity (AG) in mitigating the effects of microgravity aboard the International Space Station (ISS) on ocular structure and BRB integrity. Our proposed study will define the relationship between gravitational dose and physiological response by assessing gravity levels ranging from 0 to 1 g. Although gravitational response has been obtained for some biochemical systems in very limited animal studies, these dose–responses are unknown in the neurovascular system. We will determine the operating range of AG levels that are most likely to be effective as a countermeasure for ocular tissue. Our studies will improve our understanding of the mechanisms of neurovascular and cellular adaptation to chronically altered gravity levels. The flight low anterior resection (LAR) dissection was completed in April 2023. Both eyes from each group were collected. The left eye was fixed in PFA, and the retina and optic nerve were dissected out from the right eye and fresh frozen at -80c. Retina and optic nerve samples were directly sent to NASA GeneLab from Kennedy Space Center (KSC) for genomic analysis. Fixed right eyes were sent to Loma Linda University histology core for process and sectioning. To determine the extent to which continuous partial gravity provided by centrifugation on the ISS as a countermeasure to mitigate the detrimental effects of microgravity on the retina, SpaceX was launched on March 14, 2022, at KSC on a 35-day mission. The eyes were evaluated in 9-week-old C57BL/6 male mice after SpaceX returned from the mission using mouse the Habitat Cage Unit (HCU) in the Japan Aerospace Exploration Agency (JAXA) facility in the ISS. Mice were under ambient microgravity or in a centrifugal habitat unit at 0.33, 0.67, or 1 G gravity. Within 48 hours of landing, the spaceflight mice were sacrificed, and their eyes were removed for analysis. Ground control mice were housed in similar housing conditions as flight groups. The level of 4-hydroxynonenal (4-HNE) protein, an oxidative-specific marker for lipid peroxidation and biomarkers for blood-retina barrier integrity was measured. Apoptosis in the retina and retinal endothelial cells was quantified using terminal deoxynucleotidyltransferase dUTP nick-end labeling (TUNEL) staining. There is a relationship between gravitational dose and physiological response in the retina. Our data revealed that AG provides protection against spaceflight-induced retinal damage with dose dependence.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Articles in Peer-reviewed Journals	Kremsky I, Ali S, Stanbouly S, Holley J, Justinen S, Pecaut M, Crapo J, Mao X. "Spaceflight-induced gene expression profiles in the mouse brain are attenuated by treatment with the antioxidant BuOE." Int J Mol Sci. 2023 Sep 1;24(17):13569. https://doi.org/10.3390/ijms241713569 ; PubMed PMID: 37686374; PubMed Central PMCID: PMC10487739 , Sep-2023
Articles in Peer-reviewed Journals	Kothiyal P, Eley G, Ilangovan H, Hoadley KA, Elgart SR, Mao XW, Eslami P. "A multi-omics longitudinal study of the murine retinal response to chronic low-dose irradiation and simulated microgravity." Sci Rep. 2022 Oct 7;12:16825. https://doi.org/10.1038/s41598-022-19360-9 ; PubMed PMID: 36207342; PubMed Central PMCID: PMC9547011 , Oct-2022

Articles in Peer-reviewed Journals	Mao X, Stanbouly S, Holley J, Pecaut M, Crapo J. "Evidence of spaceflight-induced adverse effects on photoreceptors and retinal function in the mouse eye." Int J Mol Sci. 2023 Apr 17;24(8):7362. https://doi.org/10.3390/ijms24087362 ; PubMed PMID: 37108526; PubMed Central PMCID: PMC10138634 , Apr-2023
Papers from Meeting Proceedings	Mao XW, Seta Stanbouly, Vijayalakshmi Sridharan, Antiño R Allen, Marjan Boerma. "Sex difference in low-dose space radiation-induced effects of the mouse retina. " 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. , Feb-2023
Papers from Meeting Proceedings	Prachi Kothiyal, Greg Eley, Hari Ilangovan, Katherine A. Hoadley, S. Robin Elgart, Xiao W. Mao, Parastou Eslami. "A multi-omics study of the murine retinal and neural response to chronic low-dose irradiation." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 6-9, 2023. , Feb-2023